STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 9C

Description

9.**Write R program to find the following Regressions with the sample data and visualize the regressions graphically.**

c) Logistic Regression

Procedure and Overvation:

20R01A6753

The logistic Regression is a regression model in which the response variable as categorical values such as TRUE or FALSE OR 0 or 1.

It actually measures the probability of a binary response as the value of response variable based on the mathematical equation relating it with the predictor variables(Independent variables)

the general mathematical equation for logistic regression is

y=1/e^x-(a+b1x1+b2x2+b3x3+.....+bnxn)

y is response variable

a,b1,b2,.....bn are coefficients

x1,x2,x3,.....,xn are predictor variables

The function used to create the regression model is the glm()

SYNTAX:

glm() in logistic regression is glm(formula,data,family)

Formula is the symbol presenting the relationship between the variables.

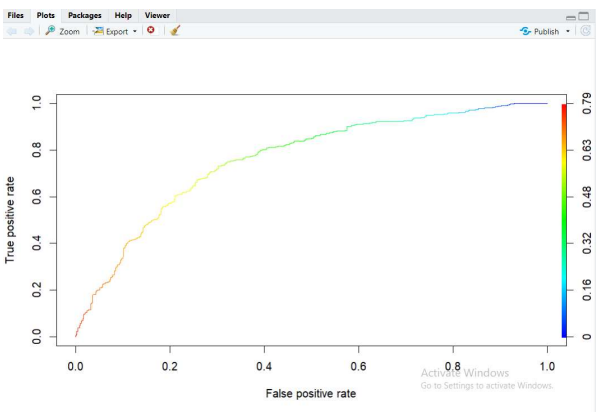
Data is the dataset giving the values of these variables.

Family is R object to specify the details of the model. Its value is binomial for logistic Regression.

SOURCE CODE:

```
claimants<-
  read.csv("C:\\Users\\student\\Desktop\\claimants.csv")
#finding null values
sum(is.na(claimants))
#removing null values na.omit(dataset)
claimants<-na.omit(claimants)
#logistic Regression
#glm(y~x,family='bin....)
logit<-glm(ATTORNEY~factor(CLMSEX)+factor(CLMINSUR)+factor(SEATBELT)
           +CLMAGE+LOSS,family="binomial",data=claimants)
summary(logit)
#confusion Matrix Table
#predict(modelobject,testdataset)
prob=predict(logit,type=c("response"),claimants)
prob
#table(dataframe1,dataframe2) to create 2x2 matrix
confusion<-table(prob>0.5,claimants$ATTORNEY)
confusion
```

```
#Model Accuracy

#adding diagonal elements in the confusion matrix

Accuracy<-sum(diag(confusion))/sum(confusion)

Accuracy

#ROC Curve

#Extract from the fitted model object the vector of filled probabilities

install.packages("ROCR")

install.packages("pROC")

library(ROCR)

library(pROC)

#prediction(probability values from model,y variables)

rocrpred<-prediction(prob,claimants$ATTORNEY)

rocrperf<-performance(rocrpred,'tpr','fpr')

plot(rocrperf,colorize=T,text.adj=c(-0.2,1.7))

#plot(rocrperf)

auc<-auc(claimants$ATTORNEY~prob)

auc
```



VIVA RESPONSES

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